Intro to ML Engineering and Linear Regression: Pre-Reads (Elaborated)

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1. Introduction to Machine Learning

What is ML?

Machine Learning is a way to **build systems that learn patterns from data instead of being explicitly programmed with rules**.

Instead of writing:

> "If user clicked 3 times → show product A"

You let the system learn:

> "Users with similar behavior tend to like product A"

Real Example

- In Netflix:
  - You are not manually assigned shows.
  - ML models analyze your watch history, time spent, genres → predict what you might like.
- In Google:
  - Search ranking is not rule-based.
  - It learns from billions of queries and clicks.

Why ML Matters

- Handles **complex patterns** humans can't manually encode
- Scales to **millions of users**
- Continuously improves with more data

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2. ML vs Classical Programming

Core Difference

Aspect	Classical Programming	Machine Learning
Input	Rules + Data	Data + Labels
Logic	Hand-written	Learned
Output	Deterministic	Probabilistic

Example: Spam Detection

Classical Approach

```
if "free money" in email:
  spam = True
```

ML Approach

- Input: Thousands of emails labeled spam/not spam
- Model learns:
    - Word patterns
    - Frequency
    - Context

Key Insight

ML is useful when:

- Rules are **too many** (e.g., fraud detection)
- Rules are **unknown** (e.g., image recognition)

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3. SDE Pipeline (Traditional Software Engineering)

Lifecycle

Requirement → Design → Code → Test → Deploy → Maintain

Example: Payment System

- Requirement: Process UPI payments
- Code: Business logic written manually
- Test: Edge cases (failed transactions)
- Deploy: Push to servers

Limitation for ML Problems

- You **can't write rules** for:
    - "What makes a good movie recommendation?"
    - "What is a cat in an image?"

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4. ML Pipeline (What Changes)

ML pipeline is **data + code + model combined**.

End-to-End Flow

Problem → Data → Features → Model → Evaluation → Deployment → Monitoring

Real Example: House Price Prediction

- Problem: Predict price
- Data: Location, size, bedrooms
- Features: price per sq ft, locality score
- Model: Linear Regression
- Output: Predicted price

Key Difference vs SDE

SDE	ML
Code-centric	Data-centric
Static	Continuously evolving
Debug code	Debug data + model

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5. Phases of Machine Learning (With Reality)

1. Data Collection

- Sources:
    - Databases
    - APIs
    - Logs
- Example: E-commerce collecting clickstream data

2. Data Preparation (Most Important)

- Missing values
- Outliers
- Encoding categorical variables

Example:

City: Bangalore → [1,0,0]

City: Delhi → [0,1,0]

3. Modeling

- Choose algorithm
- Train model on data

4. Evaluation

- Check performance on unseen data

5. Deployment

- Serve model via API

6. Monitoring

- Detect:
  - Data drift
  - Model degradation

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6. ML Categorization

ML can be categorized based on:

1. Data Labeling

- Labeled → Supervised
- Unlabeled → Unsupervised

2. Feedback Loop

- With rewards → Reinforcement Learning

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7. Types of Learning

1. Supervised Learning

- Learn mapping:
  $X \rightarrow y$

Example:

- Input: Resume
- Output: Selected / Rejected

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2. Unsupervised Learning

- No labels
- Find hidden structure

Example:

- Customer segmentation:
  - Group users into clusters (high spenders, low spenders)

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3. Reinforcement Learning

- Agent interacts with environment
- Gets rewards/penalties

Example:

- Self-driving car:
  - Reward: staying in lane
  - Penalty: collision

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8. ML Task Types (With Examples)

Regression

- Output: Continuous value
- Example:
  - Predict house price → ₹75 lakhs

Classification

- Output: Categories

- Example:
  - Spam vs Not Spam

Clustering

- Group similar data
- Example:
  - Segment customers into 5 groups

Recommendation Systems

- Suggest items

Example:

- Amazon:
  - “People who bought this also bought…”

Time Series

- Predict future from past

Example:

- Stock prices
- Demand forecasting

Reinforcement Learning Tasks

- Sequential decisions

Example:

- Game AI (e.g., chess engines)

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9. Linear Regression (Core Concept)

What It Does

Predicts a continuous value using a **linear relationship**.

Example:

- Predict salary from years of experience

Mathematical Representation

y=w1x1+w2x2+⋯+by = w_1 x_1 + w_2 x_2 + \dots + by=w1x1+w2x2+⋯+b

Meaning of Terms

- xxx: Inputs (features)
- www: Weights (importance)
- bbb: Bias (offset)
- yyy: Prediction

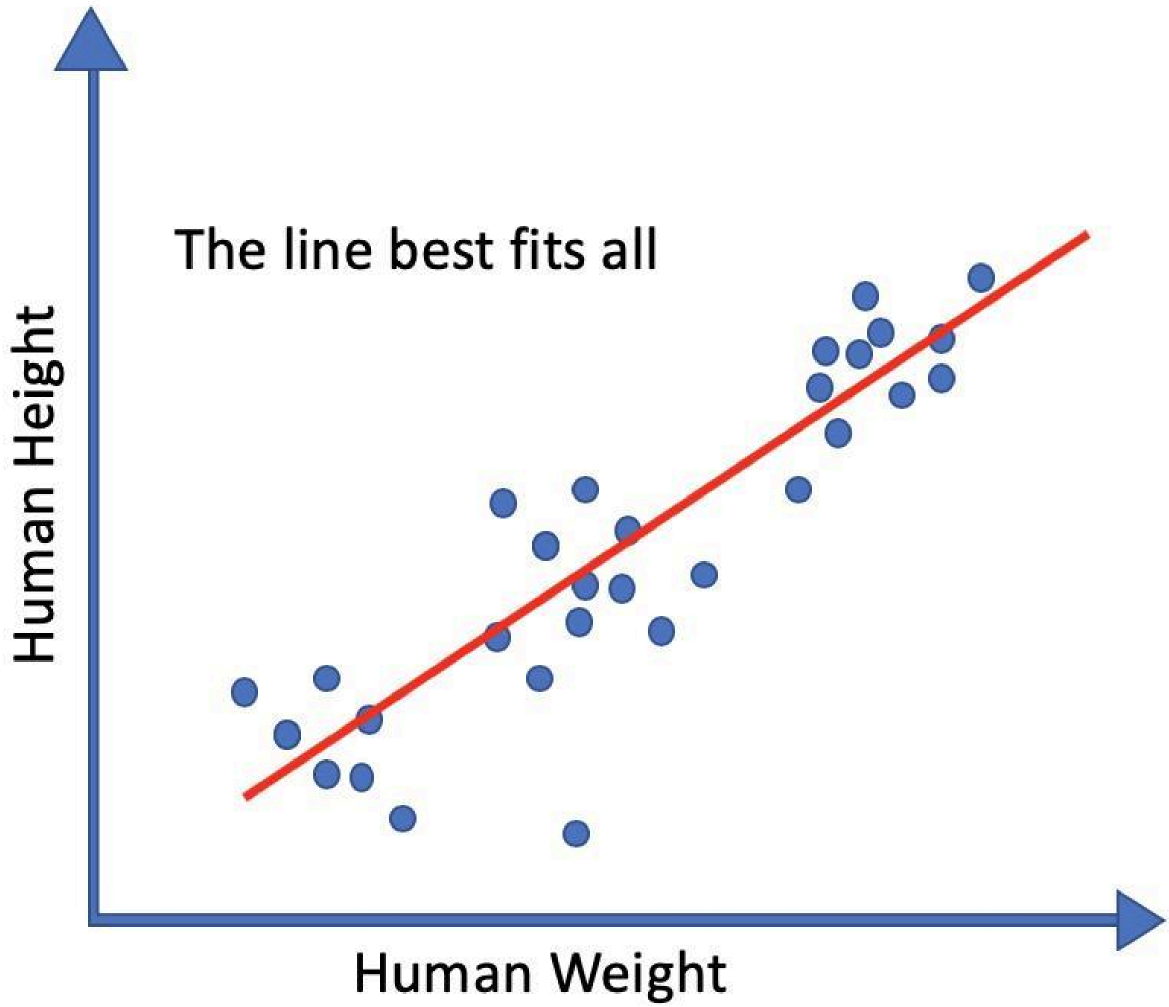
Simple Example

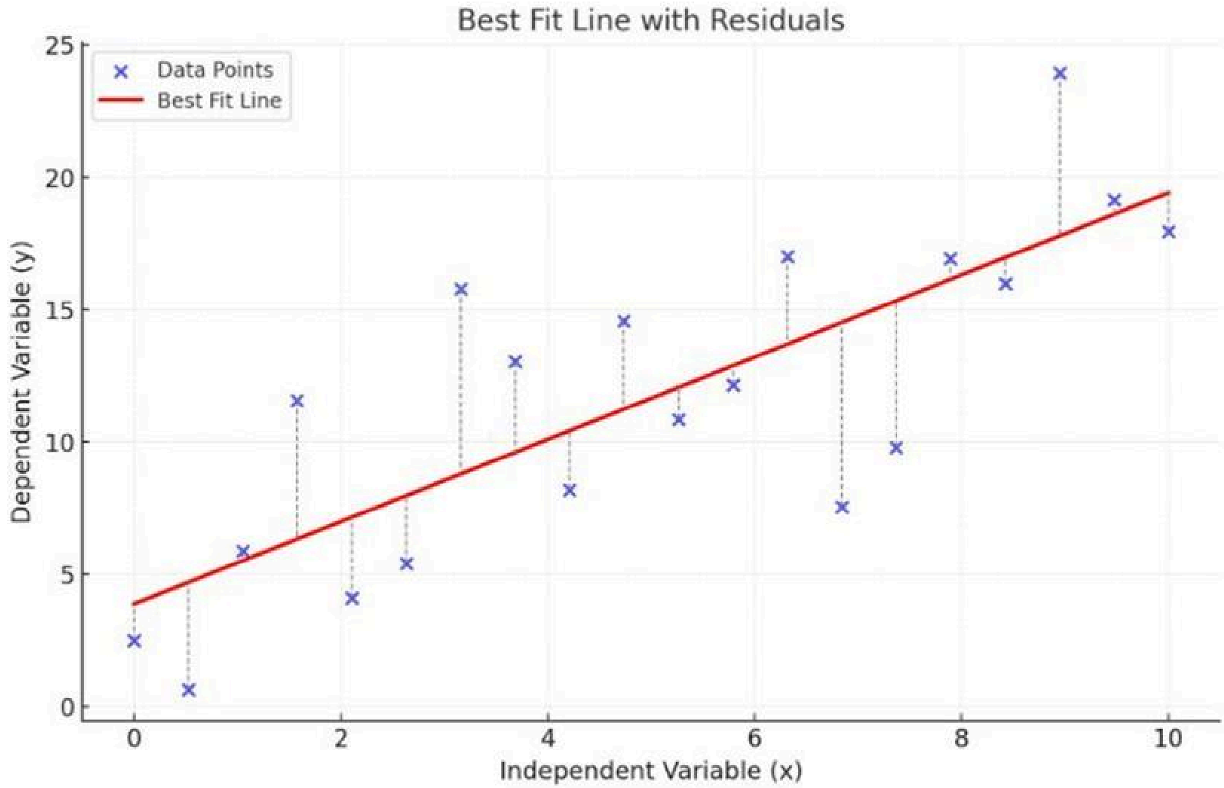
Experience (years)	Salary (₹ LPA)
1	4
2	6
3	8

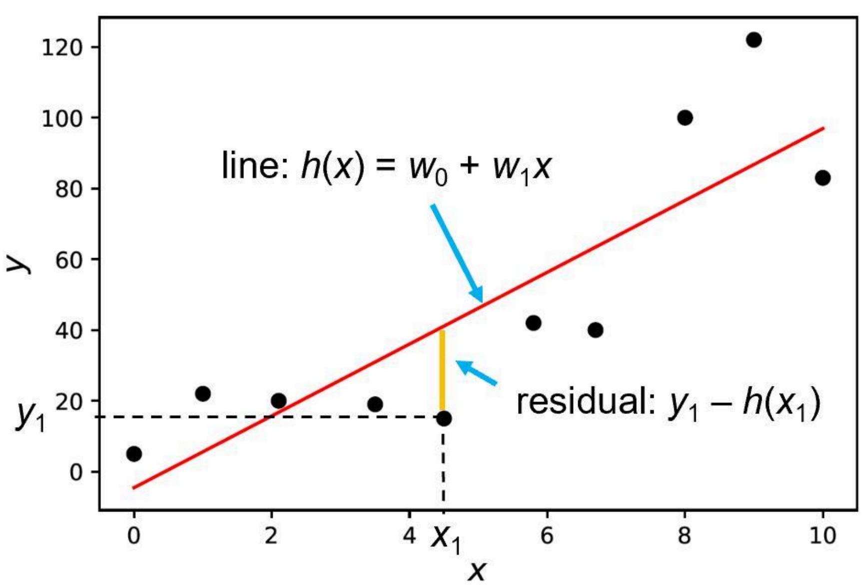
Model learns:

Salary ≈ 2 × Experience + 2

Geometric Intuition







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- In 2D → line
- In 3D → plane

- Higher dimensions → hyperplane

Why It Matters

- Easy to interpret
- Fast to train
- Foundation for:
  - Logistic regression
  - Neural networks (linear layers)

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10. Problem Statement (Critical Step)

Good Problem Statement Includes:

- Input (X)
- Output (y)
- Constraints
- Success metric

Example

> Predict delivery time based on distance, traffic, and weather.

Bad:

> “Improve delivery system”

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11. Data Notation (Standard Language)

Symbol	Meaning
X	Input features

y	Target
m	Number of samples
n	Number of features

Example:

- Dataset:
  - 100 houses → m = 100
  - 3 features → n = 3

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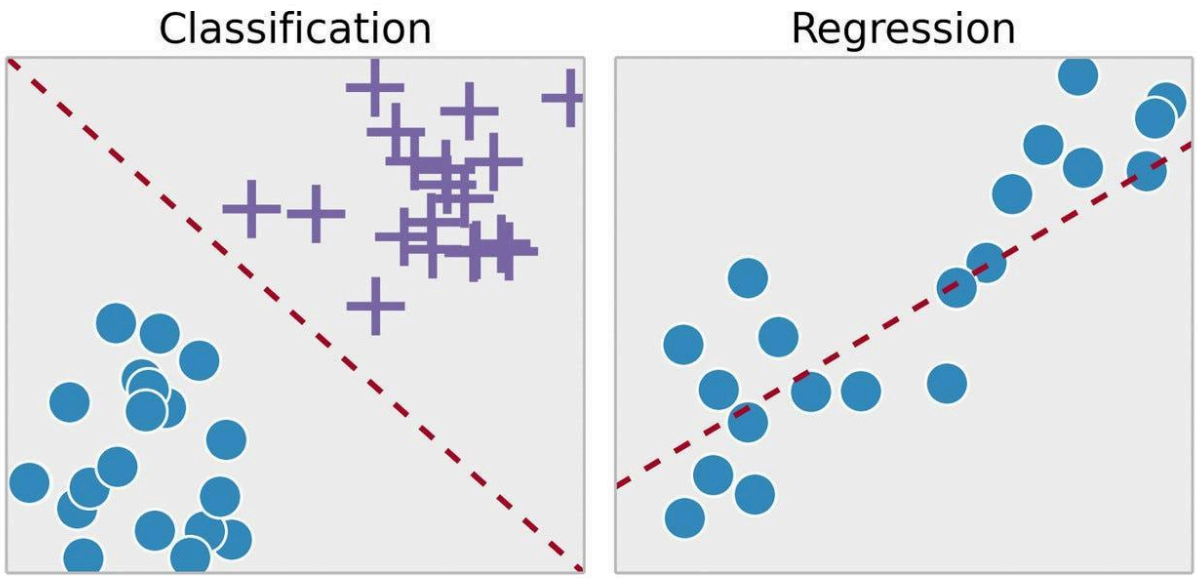
12. Geometric Intuition (Why It Helps)

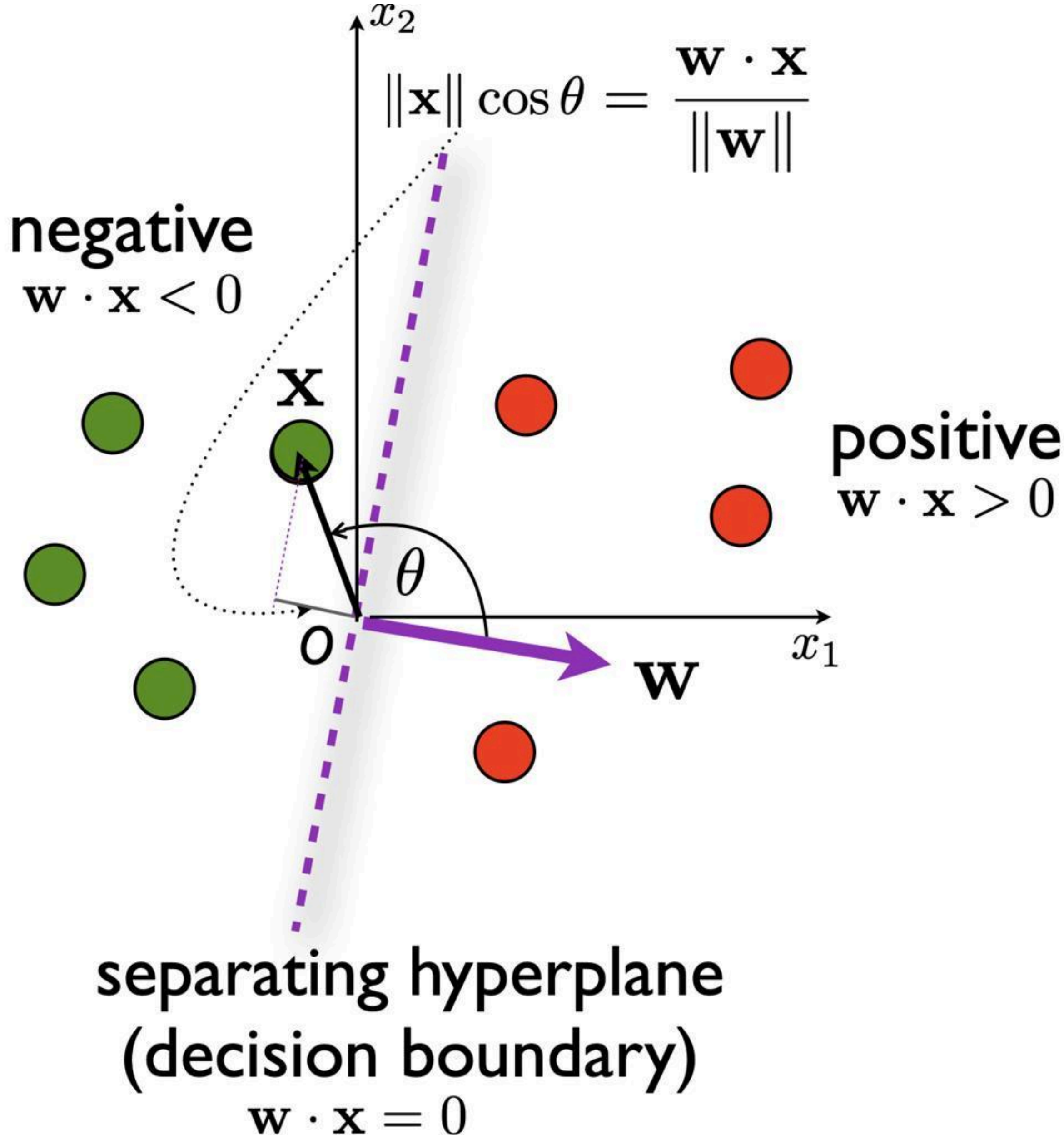
Think Visually

- Regression → fitting a line
- Classification → separating data with a boundary
- Clustering → grouping nearby points

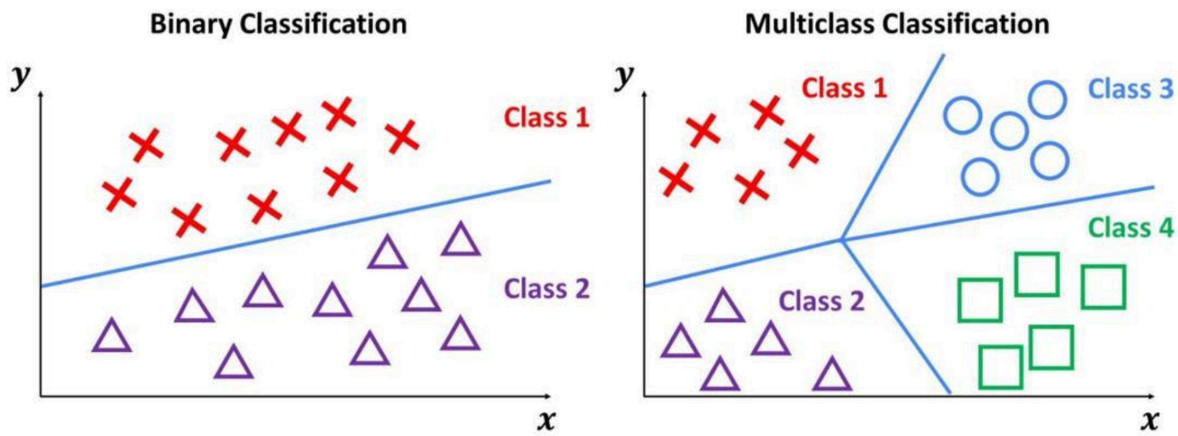
Example

- Spam vs Not Spam:
  - Model draws a **decision boundary**


Classification
Regression



x_2
$\|\mathbf{x}\| \cos\theta = \dfrac{\mathbf{w} \cdot \mathbf{x}}{\|\mathbf{w}\|}$
negative
$\mathbf{w} \cdot \mathbf{x} < 0$
$\mathbf{x}$
positive
$\mathbf{w} \cdot \mathbf{x} > 0$
θ
O
x_1
$\mathbf{w}$
separating hyperplane
(decision boundary)
$\mathbf{w} \cdot \mathbf{x} = 0$



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Final Takeaways

- ML is **data-driven decision making**
- ML pipeline ≠ traditional software pipeline
- Linear regression is the **starting point for all ML understanding**
- Real-world ML is:
  - 70% data
  - 20% engineering
  - 10% modeling